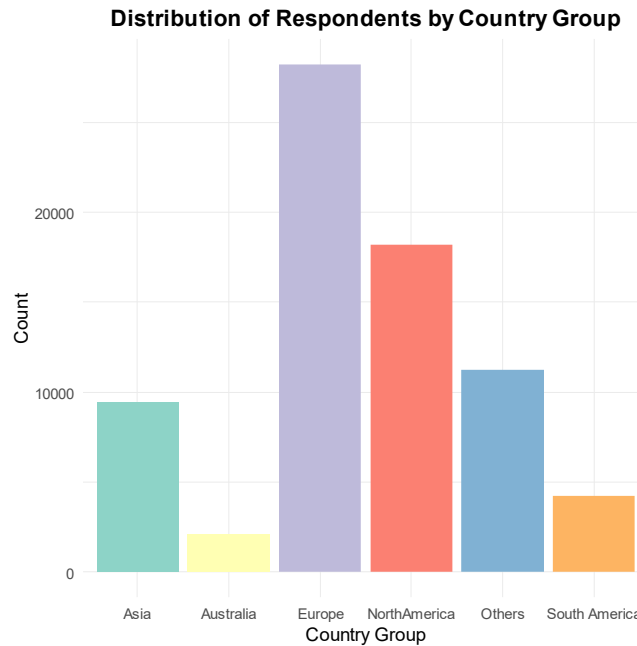


Aryan Sharma

Employment Status among Tech Professionals

Exploratory Data Analysis & Processing

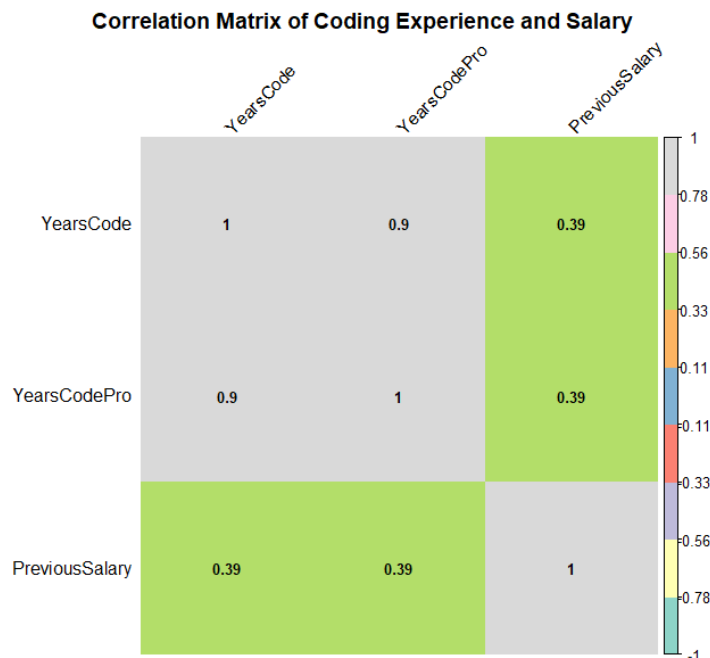
The dataset consists of 15 variables and 73,462 observations. Several variables were in character format, so they were converted into factor variables to ensure compatibility with logistic regression modeling. Examples include Employment, Employed, and MentalHealth which were transformed into categorical factors. Additionally, a new derived column named CountryGroup was created based on the existing Country variable, grouping countries into broader regions such as North America, Europe, Asia, Australia, and South America for more meaningful geographic analysis. Correlation analysis was performed among numeric variables such as YearsCode, YearsCodePro, and PreviousSalary to examine potential relationships. The dataset was then split into training (75%) and testing (25%) subsets to build and evaluate logistic regression models predicting an individual's employment status based on demographic, skill-based, and regional factors.



Correlation:

There are only 3 variables in the data which are numeric. Hence, we are calculating whether they have any correlation between them or not.

```
correlation_matrix <- cor(data[c("SeniorCitizen", "tenure", "MonthlyCharges", "TotalCharges")])
corrplot(correlation_matrix, method = "pie")
print(correlation_matrix)
```



There is a correlation between YearsCode and YearsCodePro, hence we are only taking YearsCodePro the variables.

Predictor table:

Predictor	Effects	Rationale
Gender	+/-	Gender may influence hiring outcomes due to workplace or regional biases
Employment	+	Previously employed individuals are more likely to get hired again
MentalHealth	-	Poor mental health may reduce job performance and hiring chances
MainBranch	+/-	Developers and non-developers face different job market demands
YearsCodePro	+	More coding experience increases employability as it reflects more heads on experience
PreviousSalary	+	Higher past salary indicates stronger expertise and job readiness
CountryGroup	+	Regional job market differences affect employment outcomes
ComputerSkills	+	If the tech support is good and solve all the customer related problems then the churn rate would be less

Regression Analysis

We have taken 2 models for this analysis, Model 1 and Model 2 take Logistic regression analysis approach.

```
model1 = glm(Employed ~ Gender + Employment + MentalHealth + MainBranch + YearsCodePro + CountryGroup + PreviousSalary + ComputerSkills, data=train_data, family=binomial (Link="Logit"))
```

```
model2 = glm(Employed ~ Gender + Employment + MentalHealth + MainBranch + YearsCodePro*PreviousSalary + CountryGroup + ComputerSkills, data=train_data, family=binomial (Link="Logit"))
```

Comparison of Models

	Dependent variable:	
	Employed	
	(1)	(2)
GenderNonBinary	-0.092 (0.073)	-0.091 (0.073)
GenderWoman	-0.099** (0.045)	-0.099** (0.045)
Employment1	0.140*** (0.031)	0.142*** (0.031)
MentalHealthyes	0.014 (0.024)	0.014 (0.024)
MainBranchNotDev	-0.619*** (0.038)	-0.620*** (0.038)
YearsCodePro	0.016*** (0.001)	0.010*** (0.002)
CountryGroupAustralia	0.409*** (0.065)	0.426*** (0.065)
CountryGroupEurope	0.259*** (0.033)	0.277*** (0.033)
CountryGroupNorthAmerica	0.454*** (0.040)	0.467*** (0.040)
CountryGroupothers	0.157*** (0.037)	0.167*** (0.037)
CountryGroupSouth America	-0.069 (0.049)	-0.057 (0.049)
PreviousSalary	-0.00000*** (0.00000)	-0.00001*** (0.00000)
Computerskills	0.330*** (0.003)	0.330*** (0.003)
YearsCodePro:PreviousSalary		0.00000*** (0.00000)
Constant	-4.136*** (0.052)	-4.099*** (0.053)
observations	73,462	73,462
Log Likelihood	-32,665.040	-32,659.620
Akaike Inf. Crit.	65,358.080	65,349.250
Note:	*p<0.1; **p<0.05; ***p<0.01	

Interpretation

1. Coefficient Interpretation

Variable	Interpretation
GenderWoman	OR $\approx$ 0.90 $\rightarrow$ Women have about 10% lower odds of being employed than men
Employment1	OR $\approx$ 1.15 $\rightarrow$ Currently employed people are 15% more likely to be employed again
MainBranchNotDev	OR $\approx$ 0.54 $\rightarrow$ Non-developers have 46% lower odds of employment
YearsCodePro	OR $\approx$ 1.01 $\rightarrow$ Each additional year of professional coding increases employment odds slightly
ComputerSkills	OR $\approx$ 1.39 $\rightarrow$ Strong positive effect; higher skills $\rightarrow$ higher employment odds
CountryGroupNorthAmerica	OR $\approx$ 1.57 $\rightarrow$ Respondents from North America have ~57% higher odds vs. the baseline region
CountryGroupEurope	OR $\approx$ 1.29 $\rightarrow$ Europeans also have better odds than baseline (likely Asia or missing group)
PreviousSalary	OR $\approx$ ~1.00 $\rightarrow$ negligible per-unit effect (salary scale is likely large)

2. F1 Score and Precision/Recall

Predictor	Precision Score	Recall	F1 Score	AUC Score
Model1	0.769	0.770	0.771	0.785
Model2	0.772	0.769	0.770	0.785

Recommendations:

Based on the predictor understanding and preliminary analysis, it is recommended that MNCs focus on strengthening **skill-based hiring** and **regional inclusivity**. Emphasis should be placed on improving opportunities for **women** and **non-developer roles**, as these groups show slightly lower employment odds. Programs that enhance **technical proficiency** and **mental health support** could further improve employability outcomes. Additionally, regional training initiatives in areas with lower hiring rates may help balance global employment disparities.

Our model indicates that **technical skills**, **experience**, and **location** are the key drivers of employment success in global MNCs. Gender and mental health factors also play a smaller yet meaningful role. With an AUC of 0.78 and F1 of 0.77, the model shows strong reliability in predicting hiring outcomes.